5 Simulated antenna input and runtime behaviour

DONE | TOTAL: orig ~6d · total ~0d | completed 09 Oct 2026

GOAL

Process repeatable simulated antenna input through one TimingNode using normal runtime, domain and persistence paths.

RESULT

- Simulated antenna observations reach the normal registration path.
- Successful TimingNode OPEN/CLOSE transitions are represented in the normal committed
- Registration revoke appends REV through the same committed source stream and leaves the
- Normal manual-registration ADD is available through IF-03 with client-supplied
- Simulated-tag profiles drive the normal antenna path rather than direct registration.
- Sustained input can be measured without bypassing TimingNode ownership.
- Restart/recovery works with the same simulated input path used by automated tests.

END DEMO

- Open one TimingNode and show the committed OPEN TimingData record.
- Feed repeatable tag observations through SimulatedAntenna and show which observations
- Add manual registrations using both client-selected current time and operator-entered
- Run a short simulated-tag batch and show one representative two-tag passage reaching
- Revoke one registration through IF-03 and show durable ADD plus REV source
- Close the TimingNode and show the CLOSE record in the same source sequence.
- Inspect the runtime counters, restart SI-01 and continue using the same simulated input

DOCUMENTATION

30-UC R-80 | 41-01-SSD R-70 | 32-03-IDD R-80 | 60-SVP R-70

TOOLING / ENGINEERING ENVIRONMENT

T01 after D01 — DONE

Runtime-characterization harness and evidence tooling

Completed through Java PR...

DOCUMENTATION / DECISIONS

D01 — DONE

Runtime execution and measurement plan

Define measurement questions,...

D02 — DONE

Define OPEN/CLOSE TimingData semantics and reference mapping

Defines successful CLOSED-OPEN...

D03 — DONE

Define registration revoke semantics and public contract

Defines append-only...

D04 after D01 — DONE

Review simulated antenna/input architecture

Review the simulated...

D05 after D01 — DONE

Define internal runtime-observability architecture

Define the internal...

D06 — DONE

Link SSD architecture Needs to detailed SDD design

Completed through meta issue...

D07 — DONE

Clarify Presentation external ports in architecture diagram

Completed through meta issue...

D08 — DONE

Define manual-registration API and timestamp precision

Define normal IF-03...

D09 after D04 — DONE

Define simulation layers, simulated tags and profile control

Distinguish direct accepted-re...

APPLICATION / PRODUCT

A01 after D04 — DONE

Qualify/implement simulated antenna and tag-processing path

Requalified the simulated-inpu...

A02 after D05 A01 — DONE

Qualify/implement runtime markers and counters

Requalified and corrected on...

A03 after V01 — DONE

Measurement-driven allocation/data-access decision

V01 development-host evidence...

A04 after D02 — DONE

Commit OPEN/CLOSE through the normal TimingData path

Commit successful TimingNode...

A05 after D03 — DONE

Commit registration revoke through the normal TimingData path

Add the public node-scoped...

A06 after D08 — DONE

Implement manual-registration API, precision policy and Development Client controls

Completed through Java issue...

A07 after D09 — DONE

Implement simulated-tag profiles and Development Client batch simulation

Completed through Java issue...

VERIFICATION / TEST

V01 after D01 D04 D05 T01 A01 A02 — DONE

Single-node baseline load/burst characterization

Development-host baseline is...

V02 after A03 — DONE

Sustained tag-observation load and stalled-downstream fairness/backpressure

Completed through Java issue...

V03 after A01 — DONE

Restart and recovery with simulated input

Completed through Java PR...

V04 after A01 — DONE

Provider bootstrap verification

Verified built-in EventDataPro...

V05 after A04 — DONE

OPEN/CLOSE TimingData ordering, persistence and rejection verification

Completed through Java PR...

V06 after A05 — DONE

Registration revoke/API/persistence verification (client UI evidence in Step 6 V01)

SI-01 revoke/API/persistence...

V07 after A06 — DONE

Manual registration, timestamp precision and Development Client projection verification

Completed with the automated...

V08 after A07 — DONE

Simulated-tag profile and batch-driver verification

Completed through Java issue...

PLANNING CHANGES

- 2026-10-09 — Complete Step 5 on the proven SI-01/server verification baseline. V06 closes on revoke/API/persistence evidence; GUI revoke/DELETED/reconnect/recovery verification moves to Step 6 V01 using SI-02.
- 2026-10-07 — Add D08/A06/V07 for normal manual-registration semantics and time precision, and D09/A07/V08 for deterministic simulated-tag profiles and batch control.
- 2026-10-07 — Move selected-target runtime evidence to Step 9. Step-5 V01 is the development-host baseline and releases the A03 runtime design decision.
- 2026-10-07 — Complete T01 through Java PR #327 and make V01 runtime baseline characterization the next activity.

PLANNING CHANGES (CONT.)

- 2026-10-06 — Requalify A01 on Java main and complete V04 provider bootstrap verification through PR #302 with built-in reference providers and an external provider JAR.
- 2026-10-04 — Complete D05 runtime-measurement design while keeping A01 active until the antenna/tag-processing path is wired into runtime composition.
- 2026-10-04 — Reopen D01 and decompose the work into D04/D05 design gates plus T01 harness; A01/A02 require requalification and V01 moves before A03.
- 2026-10-04 — Fix the single-TimingNode runtime-measurement baseline in D01 and make A01 the active implementation activity.

PLANNING CHANGES (CONT.)

- 2026-10-04 — Activate Step 5 after Step-4 VC-ST1-003 closure with D01 as the first planning/design activity.